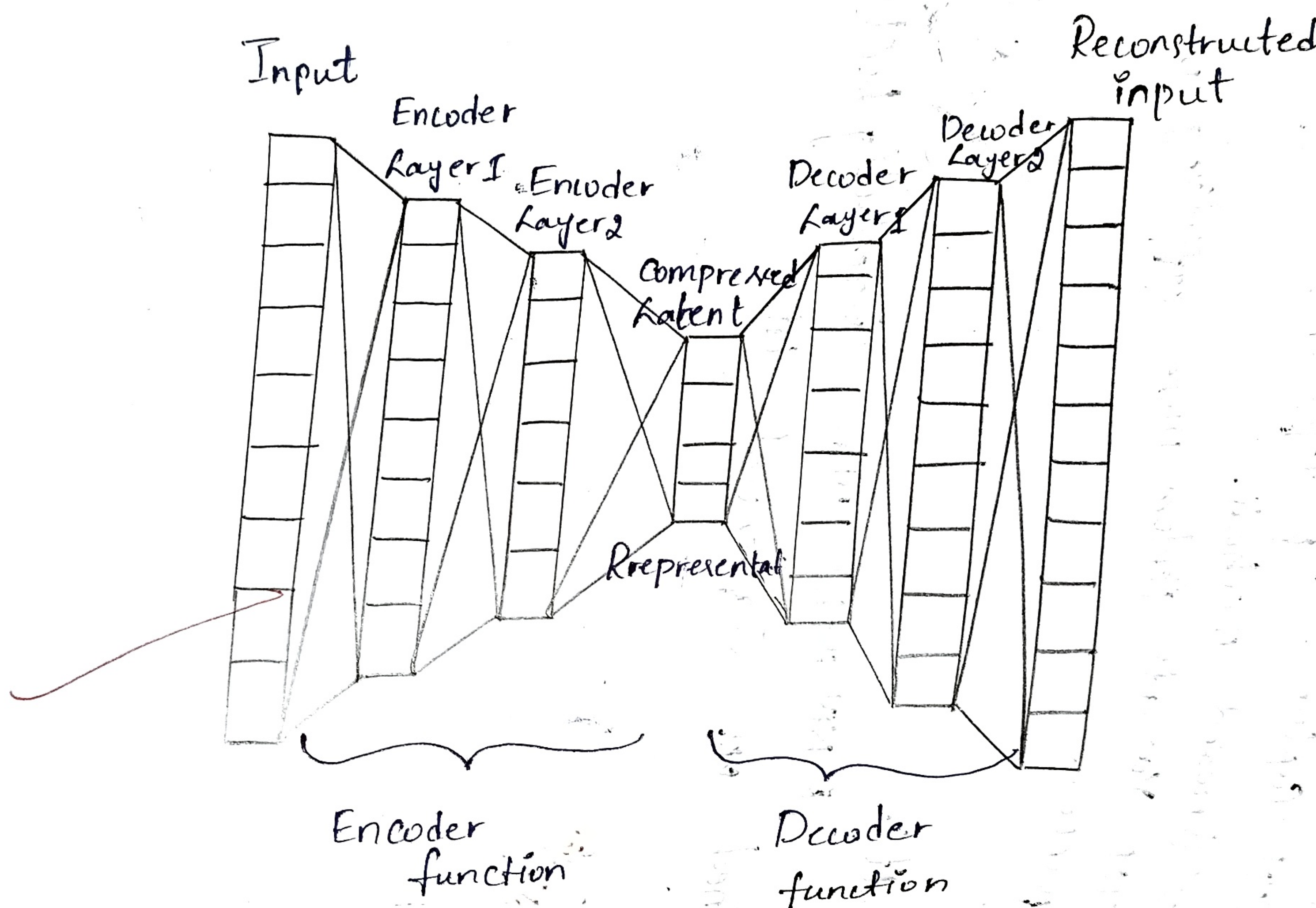
Autoencoder Architechure :-



Lab: 10 Perform Compression on mnist dataset Using autoencoder

Aim:

Design, train and evaluate a simple auto encoder to compress and reconstruct MNIST images Measure reconstruction quality ~~and show em~~

Key Ideas:

* Encoder compresses 28x28 → low-dimensional latent vector.
* Decoder reconstructs from latent back to 28x28
* Evaluate with MSE and visual inspection

Architecture:

* → Encoder: input 28x28 (flatten or Conv)
  * Dense path : 784 → 512 → 256 → latent_dim
  * or Conv path : Conv(32,3) → ReLu → Maxpool → Conv(64,3) → ReLu → Flatten → Dense (latent_dim)
  * Decoder: mirror of encoder (latent_dim → 256 → 512 → 784 → reshape 28x28)
* Activation: final output Sigmoid

Observation:

Epoch 1 : Avg Loss ~~162~~ 162.229

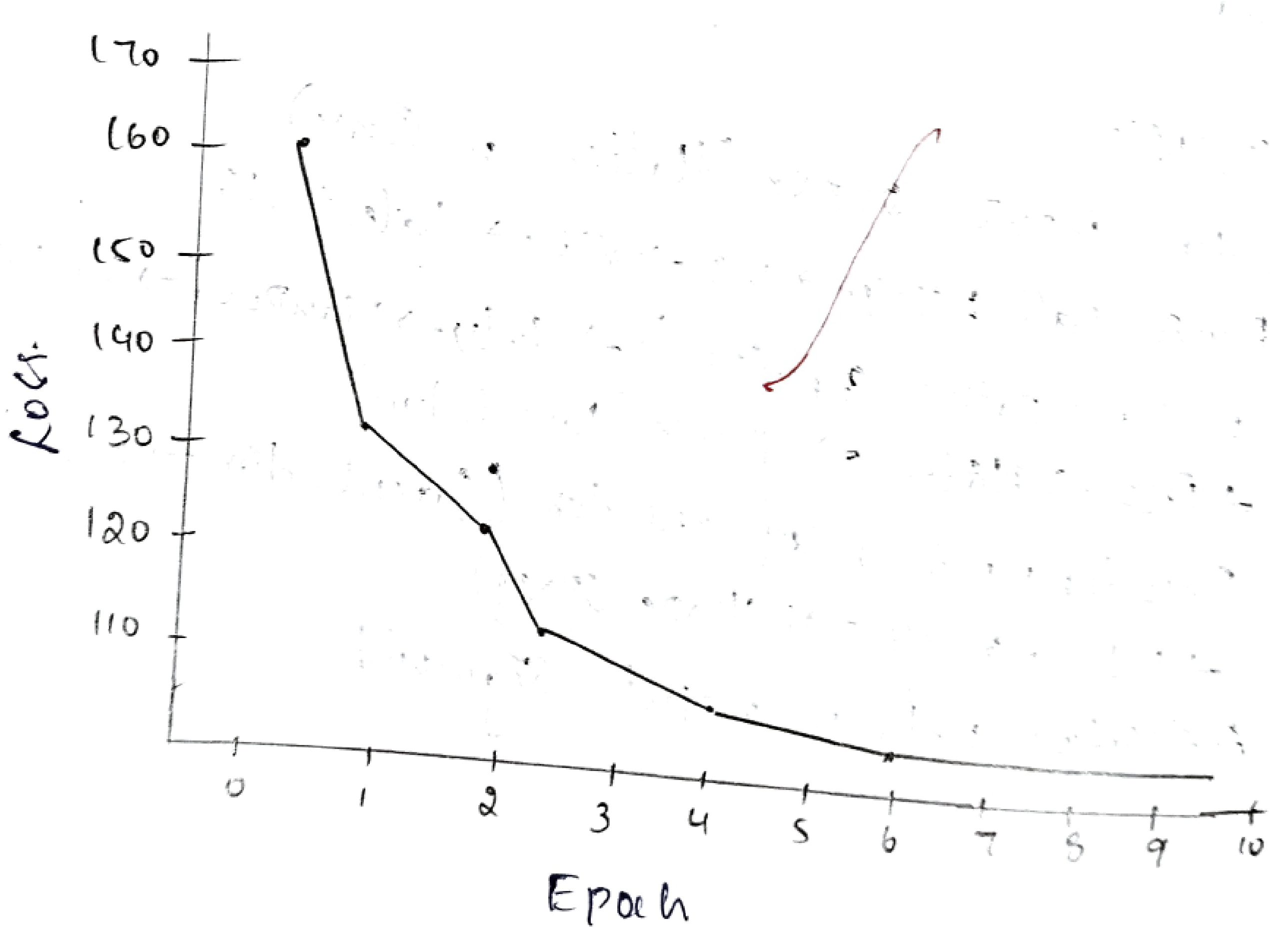
Epoch 2: Avg Loss 124.9240

Epoch 3: Avg Loss 119.5600

Epoch 4: Avg Loss 116.8461

⋮

Epoch 10 Avg Loss 111.1155.



Results:

Successfully compressed on MNIST dataset using autoencoders.

Observation:

→ The Training accuracy increases with epoch while the loss decreases.

→ The reconstructed images visually resemble the original MNIST.

→ The decreasing trend in the overall loss suggest the latent space is becoming more structured and aligned.

→ The AE successfully reconstructed the original input data from its compressed latent